

Teaching Statement

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I have been the instructor of record for eleven semesters at the University of South Carolina since Spring 2022, across seven courses spanning the introductory curriculum: Calculus I and Calculus II, Calculus Workshops, Calculus for Business Administration, Finite Mathematics, Precalculus Mathematics, and Basic and Intensive Basic College Mathematics. In each I was responsible for the full course including lecturing, writing and grading all quizzes and examinations, and setting the pace and structure of the term. I also led Calculus I recitations for three semesters before that. My CV lists the courses and terms. Nearly all of my students are in a required course they did not choose. My aim is that they notice they can do mathematics that defeated them in week one, and that they approach the next problem without deciding in advance that they cannot.

Structure: Problem Sheets and Guided Problem Solving

Unfortunately, for my first several semesters I taught from handwritten notes presented from my iPad. Students spent the class period copying. When I looked at what they had written down, it was frequently incomplete or wrong, and the transcription was consuming attention that should have gone to the mathematics. Since Fall 2024, I write a typed problem sheet for each section of the text: the definitions and formulas are stated cleanly at the top in my own words, followed by a graded sequence of problems. Students receive the sheet at the start of class and we work through it together where I present the method, do one or more to explain what to do, then set them loose on one or two problems for several minutes before we resolve it as a group, and frequently I supply the first two steps and let them finish. The change in engagement was immediate and significant, and it is one of the most effective adjustment I have made to my teaching.

Word problems required a further change. My experience has been students struggle with word problems more than any other kind of exercise, and the reason is usually not the algebra but the translation: they cannot tell what is being asked. So I now present them in clusters. One problem is worked in full, and two more follow with the same underlying structure but a different story with different objects, different quantities, same relationship to be extracted. Solving all three forces students to notice what stays fixed across the surface changes, which is the actual skill. Once students identify the structure, they can direct their attention to the algebraic work rather than to guessing what the problem is asking. Student feedback repeatedly identifies the problem sheets as a particularly useful feature of the course, both for the in-class activities and because the completed sheets become a reference for exams.

Assessment: Frequent, Low-stakes

I offer weekly quizzes as I support frequent and short formative assessments. These assessments force students to engage with material while it is still current rather than in the two days before an examination, and they give me a weekly signal about what has and has not landed. Quizzes are preceded by an in-class review session, and I write practice examinations and study guides for each unit.

Before some quiz, depending on the material, I ask the class whether they would prefer to take it individually or in pairs, and if the class is broadly uneasy about the material we do it collaboratively. This has been noted in my students' evaluations. One wrote that they had never seen an instructor offer the choice, and it has a pedagogical justification beyond goodwill: a group quiz on a shaky topic

produces the mathematical conversation I want anyway, and it tells me, from the vote itself, how confident the class actually feels. Although I am versatile in my formative assessments like quizzes, summative assessments in the form of longer examinations are still upheld and remain individual.

Responding to Feedback

I greatly value feedback. Midway through each term I distribute a one-page evaluation asking students about pace, the balance of theory and problem solving, the difficulty of quizzes and examinations, and the length of weekly assignments. I read the responses and say publicly what I am changing in the second half of the term. I also use active learning strategies like the “muddiest point”: students anonymously write down the idea that remained unclear, and I open the next class with whatever was named most often.

In addition to receiving feedback on teaching methods, I use feedback on assessments. My quiz policy is a direct product of this. I have been offering quizzes on the last meeting day of the week, and I had been including everything covered that week. This meant students were tested on material introduced the previous day. They told me, in the mid-semester forms, that this was the hardest thing about the course. I now cut the quiz off two days before it is given, and whatever is covered in between appears on the following week’s quiz instead, giving students a buffer time. The same students said the quizzes were too long, so I reduced them to two problems, or three when the class votes to work in pairs. Both changes cost nothing and removed a real obstacle. These changes have been largely appreciated and there is concrete evidence supporting that. Between Fall 2025 and Spring 2026 my end-of-term evaluations improved on every item, on response rates near 85%. Overall instructor effectiveness rose from 3.77 to 4.03 out of 5, the item on creating an environment in which students feel encouraged to participate rose from 4.13 to 4.48, and organization of course materials rose from 4.03 to 4.30.

Supporting Students and Make an Inclusive Classroom

Not every obstacle to learning is mathematical. In one of my courses at USC, a student was on track to fail as their health condition made the course schedule impossible for them to follow. Although exam accommodations were arranged through USC’s Student Disability Resource Center, the quiz days and assignment deadlines were within the instructors discretion. I made sure the student was able to complete all requirements with adjusted deadlines. The student was not in the right condition to take the final exam. Instead of failing them, I assigned them an incomplete grade on the final exam and they took the exam early next semester. That year, I was nominated for a Thank a Teacher Certificate of Appreciation from USC’s Center for Teaching Excellence, which I value more than any other recognition I have received.

The same flexibility is available to everyone. I announce the makeup policy early, so asking does not feel like asking a favor. I post a PDF of all board work as I lecture on an iPad so understanding the lecture does not depend on catching every word. I check comprehension by asking students to restate a step rather than by asking whether everyone follows. And I work live at the board, which means I make mistakes and when a student catches one I thank them, which shows that correctness comes from checking, not authority.

Mentoring First-Time Instructors

In Spring 2023 I applied to and was selected for an NSF-funded peer-mentoring program in my department, which pairs experienced graduate instructors with graduate students preparing to be an

instructor of record for the first time. Throughout the year, I mentored seven graduate students through that transition. I observed each of them three times per semester which included watching the lecture itself, the classroom management, and how students responded. These observations were followed with written and verbal feedback. I also ran biweekly small-group meetings across the year, covering syllabus design, examination length, grading schemes, student engagement, and the difficulties that first-time instructors reliably hit in their first month: pacing a fifty-minute class, holding a room that has stopped listening, and deciding how much help is too much. Mentoring seven people over a year taught me to diagnose a teaching problem from the outside and to say what I saw in a way the person could act on, which is a different skill from teaching well myself.

Outlook

Teaching these courses has changed how I understand the mathematics I do myself: every time I build an explanation from the ground up for someone who does not yet see why the question is interesting, I understand the object better than I did. The two aims I began with are what the rest of it is built around. Weekly quizzes and worked problem sheets exist so that progress is visible to the student and not only to me. A student who could not factor a quadratic in September and can do so in November and can notice the point in the semester it changed. The group-quiz vote, the makeup policy, and my own mistakes at the board exist so that attempting a problem costs nothing when it goes wrong. It is also where I most often hear a student say "I am just bad at math." I tell them that no one is bad at math, that we are all still learning, and that the goal, as a high school teacher of mine once put it, is to absorb the ideas rather than adsorb them — to take them in rather than let them cling to the surface.